

Question	Marking Guidance	Mark	Comments
09.1	1. Random samples; 2. Large sample size;	2 (2 x AO3)	1. Allow in context of fish or gills 2. If a specified number is given, it must be 10 or more. 2. Accept 'many'/'multiple' for large sample but ignore 'several'

Question	Marking Guidance	Mark	Comments
09.2	Correct answer of $18\,000/1.8 \times 10^4 = 2$ marks;; 1 mark for Correct answers not given to 2sf (17 969.952) OR Evidence of 1701.7 (total length of filaments) OR 27 227.2 – 27370 (total number of lamellae)	2 (2 x AO2)	Accept 17970/17969.9

Question	Marking Guidance	Mark	Comments
09.3	(<i>Trachurus trachurus</i>) – no mark 1. More oxygen uptake/diffusion OR More gas exchange; 2. More energy/ATP from respiration;	2 (2 x AO2)	Incorrect fish name = zero Accept just 'Trachurus' or 'trachurus' 'More' required once only

Question	Marking Guidance	Mark	Comments
09.4	1 mark for first column 1 mark for 2nd and 3rd column all correct		
	Kingdom	Animalia	
	Phylum/Phyla	Chordata	
	Class	Actinopterygii	
	Order	Batrachoidiformes	
	Family	Batrachoididae	
	Genus	Opsanus	
	Species	tau	
		2 (2 x AO2)	
